



Three-tier Application Deployment on AWS EKS

Follow these steps to deploy the application on AWS EKS.

Code repository

<https://github.com/LondheShubham153/TWSThreeTierAppChallenge>

Video Tutorial

https://youtube.com/live/wgmYbSN6_Is

IAM

Create a user “eks-admin” with AdministratorAccess

Create Security Credentials Access Key and Secret access key

EC2

Create an ubuntu instance (region us-west-2)

ssh to the instance from local

Install AWS CLI v2

```
curl "https://awscli.amazonaws.com/awscli-exe-linux-x86_64.zip" -o "awscliv2.zip"
sudo apt install unzip
unzip awscliv2.zip
sudo ./aws/install -i /usr/local/aws-cli -b /usr/local/bin --update
```

Setup your access by

aws configure

Install Docker

```
sudo apt-get update
sudo apt install docker.io
docker ps
sudo chown $USER /var/run/docker.sock
```

Install kubectl

```
curl -o kubectl
https://amazon-eks.s3.us-west-2.amazonaws.com/1.19.6/2021-01-05/bin/linux/amd64/kubectl
chmod +x ./kubectl
sudo mv ./kubectl /usr/local/bin
kubectl version --short --client
```

Install eksctl

```
curl --silent --location "https://github.com/weaveworks/eksctl/releases/latest/download/eksctl_$(uname -s)_amd64.tar.gz" | tar xz -C /tmp
sudo mv /tmp/eksctl /usr/local/bin
eksctl version
```

Setup EKS Cluster

```
eksctl create cluster --name three-tier-cluster --region us-west-2 --node-type t2.medium
--nodes-min 2 --nodes-max 2
aws eks update-kubeconfig --region us-west-2 --name three-tier-cluster
kubectl get nodes
```

Run Manifests

```
kubectl create namespace two-tier-ns
kubectl apply -f .
Kubectl delete -f .
```

```
eksctl delete cluster --name my-cluster --region us-west-2
```

Install AWS Load Balancer

```
curl -O
https://raw.githubusercontent.com/kubernetes-sigs/aws-load-balancer-controller/v2.5.4/docs/install/iam\_policy.json
```

```
aws iam create-policy --policy-name AWSLoadBalancerControllerIAMPolicy
--policy-document file:///iam_policy.json
```

```
eksctl utils associate-iam-oidc-provider --region=us-west-2 --cluster=my-cluster --approve
```

```
eksctl create iamserviceaccount --cluster=my-cluster --namespace=kube-system
--name=aws-load-balancer-controller --role-name AmazonEKSLoadBalancerControllerRole
```

```
--attach-policy-arn=arn:aws:iam::626072240565:policy/AWSLoadBalancerControllerIAMPolicy  
--approve --region=us-west-2
```

```
sudo snap install helm --classic
```

```
helm repo add eks https://aws.github.io/eks-charts
```

```
helm repo update eks
```

```
helm install aws-load-balancer-controller eks/aws-load-balancer-controller -n kube-system  
--set clusterName=my-cluster --set serviceAccount.create=false --set  
serviceAccount.name=aws-load-balancer-controller
```

```
kubectl get deployment -n kube-system aws-load-balancer-controller
```

```
kubectl apply -f full_stack_lb.yaml
```